

DETERMINANTS MCQ — SUPER FAST TRICKS

Class 12 Applied Maths — By Thunderstudy

Think of a determinant like a machine that gives one number from rows & columns. Your goal in MCQs = avoid full solving 😊

TRICK 1: Two Rows/Columns Same $\Rightarrow$ Answer = 0

👉 If any two rows OR columns are identical, determinant is **ZERO**.

$$\begin{vmatrix} 2 & 3 \\ 2 & 3 \end{vmatrix} = 0$$

★ **MCQ shortcut:** Scan rows first before solving.

TRICK 2: One Row/Column All Zeros $\Rightarrow$ Answer = 0

If any row or column has only zeros, Determinant = 0 instantly.

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{vmatrix} = 0$$

TRICK 3: Proportional Rows $\Rightarrow$ Determinant = 0

If one row is a multiple of another (e.g., $\text{Row}_2 = 2 \times \text{Row}_1$), $\text{Det} = 0$.

 Look for ratios quickly: **2 $\rightarrow$ 4, 3 $\rightarrow$ 6, 5 $\rightarrow$ 10**. Solved in 3 seconds!

TRICK 4: Swap Rows $\Rightarrow$ Sign Changes Only

If rows are interchanged, determinant becomes negative: $|A| \rightarrow -|A|$

 Useful when options differ only by sign.

TRICK 5: Triangle Shape = Multiply Diagonal

If matrix is Upper triangular OR Lower triangular, Determinant = diagonal multiplication.

$$\begin{vmatrix} 2 & 1 & 4 \\ 0 & 5 & 3 \\ 0 & 0 & 7 \end{vmatrix} = 2 \times 5 \times 7 = 70$$

 **NO expansion needed.**

TRICK 6: Factor Common Number OUT

If a row has a common factor, take it out. Takes numbers smaller → faster solving.

TRICK 7: 2×2 Determinant = Cross Multiply Rule

MOST IMPORTANT 

Formula: $ad - bc$ (Main diagonal – Other diagonal)

TRICK 8: Add/Subtract Rows (Magic Trick)

Operations like $R_2 \rightarrow R_2 - R_1$ or $C_2 \rightarrow C_2 + C_1$ DO NOT change the value. Purpose: Create zeros for easier solving.

TRICK 9: Identity Matrix Shortcut

If matrix is Identity (diagonal 1s, others 0), Determinant = **1**.

TRICK 10: Parameter MCQs (x, k etc.)

If determinant = 0, look for proportional rows after substitution. Most parameter questions secretly test this.

5-SECOND EXAM STRATEGY

1. Step 1 — Check identical rows
2. Step 2 — Check zeros row
3. Step 3 — Check proportional rows
4. Step 4 — Check triangular form
5. Step 5 — Use $ad-bc$ or diagonal multiply

 70% MCQs finish without full solving.

GOLDEN MEMORY LINE (Revise Before Exam)

"Same / Zero / Multiple $\rightarrow$ Determinant Zero
Triangle $\rightarrow$ Multiply diagonal | $2 \times 2 \rightarrow ad - bc$ "

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